



National  
Qualifications  
2018

**X857/75/11**

**Physics  
Relationships Sheet**

TUESDAY, 8 MAY  
1:00 PM – 3:30 PM

---



\* X 8 5 7 7 5 1 1 \*

$$d = vt$$

$$d = \bar{v}t$$

$$s = vt$$

$$s = \bar{v}t$$

$$a = \frac{v-u}{t}$$

$$F = ma$$

$$W = mg$$

$$E_w = Fd$$

$$E_p = mgh$$

$$E_k = \frac{1}{2}mv^2$$

$$Q = It$$

$$V = IR$$

$$V_2 = \left( \frac{R_2}{R_1 + R_2} \right) V_s$$

$$\frac{V_1}{V_2} = \frac{R_1}{R_2}$$

$$R_T = R_1 + R_2 + \dots$$

$$\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2} + \dots$$

$$P = \frac{E}{t}$$

$$P = IV$$

$$P = I^2 R$$

$$P = \frac{V^2}{R}$$

$$E_h = cm\Delta T$$

$$E_h = ml$$

$$p = \frac{F}{A}$$

$$p_1 V_1 = p_2 V_2$$

$$\frac{p_1}{T_1} = \frac{p_2}{T_2}$$

$$\frac{V_1}{T_1} = \frac{V_2}{T_2}$$

$$\frac{pV}{T} = \text{constant}$$

$$f = \frac{N}{t}$$

$$v = f\lambda$$

$$T = \frac{1}{f}$$

$$A = \frac{N}{t}$$

$$D = \frac{E}{m}$$

$$H = Dw_r$$

$$\dot{H} = \frac{H}{t}$$

# Additional Relationships

## Circle

$$\text{circumference} = 2\pi r$$

$$\text{area} = \pi r^2$$

## Sphere

$$\text{area} = 4\pi r^2$$

$$\text{volume} = \frac{4}{3}\pi r^3$$

## Trigonometry

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

$$\cos \theta = \frac{\text{adjacent}}{\text{hypotenuse}}$$

$$\tan \theta = \frac{\text{opposite}}{\text{adjacent}}$$

$$\sin^2 \theta + \cos^2 \theta = 1$$

## Electron Arrangements of Elements

Group 1      Group 2

(1)

|          |          |     |
|----------|----------|-----|
| 1        | <b>H</b> |     |
| 1        |          |     |
| Hydrogen |          | (2) |

|         |           |           |           |
|---------|-----------|-----------|-----------|
| 3       | <b>Li</b> | 4         | <b>Be</b> |
| 2,1     |           | 2,2       |           |
| Lithium |           | Beryllium |           |

|        |           |           |           |
|--------|-----------|-----------|-----------|
| 11     | <b>Na</b> | 12        | <b>Mg</b> |
| 2,8,1  |           | 2,8,2     |           |
| Sodium |           | Magnesium |           |

|           |          |         |           |
|-----------|----------|---------|-----------|
| 19        | <b>K</b> | 20      | <b>Ca</b> |
| 2,8,8,1   |          | 2,8,8,2 |           |
| Potassium |          | Calcium |           |

|            |           |            |           |
|------------|-----------|------------|-----------|
| 37         | <b>Rb</b> | 38         | <b>Sr</b> |
| 2,8,18,8,1 |           | 2,8,18,8,2 |           |
| Rubidium   |           | Strontium  |           |

|               |           |               |           |
|---------------|-----------|---------------|-----------|
| 55            | <b>Cs</b> | 56            | <b>Ba</b> |
| 2,8,18,18,8,1 |           | 2,8,18,18,8,2 |           |
| Caesium       |           | Barium        |           |

|                  |           |                  |           |
|------------------|-----------|------------------|-----------|
| 87               | <b>Fr</b> | 88               | <b>Ra</b> |
| 2,8,18,32,18,8,1 |           | 2,8,18,32,18,8,2 |           |
| Francium         |           | Radium           |           |

### Key

|                      |
|----------------------|
| Atomic number        |
| Symbol               |
| Electron arrangement |
| Name                 |

### Transition Elements

|                  |                   |                   |                   |                   |                   |                   |                   |                   |                   |
|------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| (3)              | (4)               | (5)               | (6)               | (7)               | (8)               | (9)               | (10)              | (11)              | (12)              |
| 21               | 22                | 23                | 24                | 25                | 26                | 27                | 28                | 29                | 30                |
| <b>Sc</b>        | <b>Ti</b>         | <b>V</b>          | <b>Cr</b>         | <b>Mn</b>         | <b>Fe</b>         | <b>Co</b>         | <b>Ni</b>         | <b>Cu</b>         | <b>Zn</b>         |
| 2,8,9,2          | 2,8,10,2          | 2,8,11,2          | 2,8,13,1          | 2,8,13,2          | 2,8,14,2          | 2,8,15,2          | 2,8,16,2          | 2,8,18,1          | 2,8,18,2          |
| Scandium         | Titanium          | Vanadium          | Chromium          | Manganese         | Iron              | Cobalt            | Nickel            | Copper            | Zinc              |
| 39               | 40                | 41                | 42                | 43                | 44                | 45                | 46                | 47                | 48                |
| <b>Y</b>         | <b>Zr</b>         | <b>Nb</b>         | <b>Mo</b>         | <b>Tc</b>         | <b>Ru</b>         | <b>Rh</b>         | <b>Pd</b>         | <b>Ag</b>         | <b>Cd</b>         |
| 2,8,18,9,2       | 2,8,18,10,2       | 2,8,18,12,1       | 2,8,18,13,1       | 2,8,18,13,2       | 2,8,18,15,1       | 2,8,18,16,1       | 2,8,18,18,0       | 2,8,18,18,1       | 2,8,18,18,2       |
| Yttrium          | Zirconium         | Niobium           | Molybdenum        | Technetium        | Ruthenium         | Rhodium           | Palladium         | Silver            | Cadmium           |
| 57               | 72                | 73                | 74                | 75                | 76                | 77                | 78                | 79                | 80                |
| <b>La</b>        | <b>Hf</b>         | <b>Ta</b>         | <b>W</b>          | <b>Re</b>         | <b>Os</b>         | <b>Ir</b>         | <b>Pt</b>         | <b>Au</b>         | <b>Hg</b>         |
| 2,8,18,18,9,2    | 2,8,18,32,10,2    | 2,8,18,32,11,2    | 2,8,18,32,12,2    | 2,8,18,32,13,2    | 2,8,18,32,14,2    | 2,8,18,32,15,2    | 2,8,18,32,17,1    | 2,8,18,32,18,1    | 2,8,18,32,18,2    |
| Lanthanum        | Hafnium           | Tantalum          | Tungsten          | Rhenium           | Osmium            | Iridium           | Platinum          | Gold              | Mercury           |
| 89               | 104               | 105               | 106               | 107               | 108               | 109               | 110               | 111               | 112               |
| <b>Ac</b>        | <b>Rf</b>         | <b>Db</b>         | <b>Sg</b>         | <b>Bh</b>         | <b>Hs</b>         | <b>Mt</b>         | <b>Ds</b>         | <b>Rg</b>         | <b>Cn</b>         |
| 2,8,18,32,18,9,2 | 2,8,18,32,32,10,2 | 2,8,18,32,32,11,2 | 2,8,18,32,32,12,2 | 2,8,18,32,32,13,2 | 2,8,18,32,32,14,2 | 2,8,18,32,32,15,2 | 2,8,18,32,32,17,1 | 2,8,18,32,32,18,1 | 2,8,18,32,32,18,2 |
| Actinium         | Rutherfordium     | Dubnium           | Seaborgium        | Bohrium           | Hassium           | Meitnerium        | Darmstadtium      | Roentgenium       | Copernicium       |

|                |                |                |                |                |                |  |  |  |  |
|----------------|----------------|----------------|----------------|----------------|----------------|--|--|--|--|
| 2              | <b>He</b>      |                |                |                |                |  |  |  |  |
|                | 2              |                |                |                |                |  |  |  |  |
| Helium         |                |                |                |                |                |  |  |  |  |
| (13)           | (14)           | (15)           | (16)           | (17)           |                |  |  |  |  |
| 5              | 6              | 7              | 8              | 9              | 10             |  |  |  |  |
| <b>B</b>       | <b>C</b>       | <b>N</b>       | <b>O</b>       | <b>F</b>       | <b>Ne</b>      |  |  |  |  |
| 2,3            | 2,4            | 2,5            | 2,6            | 2,7            | 2,8            |  |  |  |  |
| Boron          | Carbon         | Nitrogen       | Oxygen         | Fluorine       | Neon           |  |  |  |  |
| 13             | 14             | 15             | 16             | 17             | 18             |  |  |  |  |
| <b>Al</b>      | <b>Si</b>      | <b>P</b>       | <b>S</b>       | <b>Cl</b>      | <b>Ar</b>      |  |  |  |  |
| 2,8,3          | 2,8,4          | 2,8,5          | 2,8,6          | 2,8,7          | 2,8,8          |  |  |  |  |
| Aluminium      | Silicon        | Phosphorus     | Sulfur         | Chlorine       | Argon          |  |  |  |  |
| 31             | 32             | 33             | 34             | 35             | 36             |  |  |  |  |
| <b>Ga</b>      | <b>Ge</b>      | <b>As</b>      | <b>Se</b>      | <b>Br</b>      | <b>Kr</b>      |  |  |  |  |
| 2,8,18,3       | 2,8,18,4       | 2,8,18,5       | 2,8,18,6       | 2,8,18,7       | 2,8,18,8       |  |  |  |  |
| Gallium        | Germanium      | Arsenic        | Selenium       | Bromine        | Krypton        |  |  |  |  |
| 49             | 50             | 51             | 52             | 53             | 54             |  |  |  |  |
| <b>In</b>      | <b>Sn</b>      | <b>Sb</b>      | <b>Te</b>      | <b>I</b>       | <b>Xe</b>      |  |  |  |  |
| 2,8,18,18,3    | 2,8,18,18,4    | 2,8,18,18,5    | 2,8,18,18,6    | 2,8,18,18,7    | 2,8,18,18,8    |  |  |  |  |
| Indium         | Tin            | Antimony       | Tellurium      | Iodine         | Xenon          |  |  |  |  |
| 81             | 82             | 83             | 84             | 85             | 86             |  |  |  |  |
| <b>Tl</b>      | <b>Pb</b>      | <b>Bi</b>      | <b>Po</b>      | <b>At</b>      | <b>Rn</b>      |  |  |  |  |
| 2,8,18,32,18,3 | 2,8,18,32,18,4 | 2,8,18,32,18,5 | 2,8,18,32,18,6 | 2,8,18,32,18,7 | 2,8,18,32,18,8 |  |  |  |  |
| Thallium       | Lead           | Bismuth        | Polonium       | Astatine       | Radon          |  |  |  |  |

### Lanthanides

|               |               |               |               |               |               |               |               |               |               |               |               |               |               |               |
|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|---------------|
| 57            | 58            | 59            | 60            | 61            | 62            | 63            | 64            | 65            | 66            | 67            | 68            | 69            | 70            | 71            |
| <b>La</b>     | <b>Ce</b>     | <b>Pr</b>     | <b>Nd</b>     | <b>Pm</b>     | <b>Sm</b>     | <b>Eu</b>     | <b>Gd</b>     | <b>Tb</b>     | <b>Dy</b>     | <b>Ho</b>     | <b>Er</b>     | <b>Tm</b>     | <b>Yb</b>     | <b>Lu</b>     |
| 2,8,18,18,9,2 | 2,8,18,20,8,2 | 2,8,18,21,8,2 | 2,8,18,22,8,2 | 2,8,18,23,8,2 | 2,8,18,24,8,2 | 2,8,18,25,8,2 | 2,8,18,25,9,2 | 2,8,18,27,8,2 | 2,8,18,28,8,2 | 2,8,18,29,8,2 | 2,8,18,30,8,2 | 2,8,18,31,8,2 | 2,8,18,32,8,2 | 2,8,18,32,9,2 |
| Lanthanum     | Cerium        | Praseodymium  | Neodymium     | Promethium    | Samarium      | Europium      | Gadolinium    | Terbium       | Dysprosium    | Holmium       | Erbium        | Thulium       | Ytterbium     | Lutetium      |

### Actinides

|                  |                   |                  |                  |                  |                  |                  |                  |                  |                  |                  |                  |                  |                  |                  |
|------------------|-------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|------------------|
| 89               | 90                | 91               | 92               | 93               | 94               | 95               | 96               | 97               | 98               | 99               | 100              | 101              | 102              | 103              |
| <b>Ac</b>        | <b>Th</b>         | <b>Pa</b>        | <b>U</b>         | <b>Np</b>        | <b>Pu</b>        | <b>Am</b>        | <b>Cm</b>        | <b>Bk</b>        | <b>Cf</b>        | <b>Es</b>        | <b>Fm</b>        | <b>Md</b>        | <b>No</b>        | <b>Lr</b>        |
| 2,8,18,32,18,9,2 | 2,8,18,32,18,10,2 | 2,8,18,32,20,9,2 | 2,8,18,32,21,9,2 | 2,8,18,32,22,9,2 | 2,8,18,32,24,8,2 | 2,8,18,32,25,9,2 | 2,8,18,32,25,9,2 | 2,8,18,32,27,8,2 | 2,8,18,32,28,8,2 | 2,8,18,32,29,8,2 | 2,8,18,32,30,8,2 | 2,8,18,32,31,8,2 | 2,8,18,32,32,8,2 | 2,8,18,32,32,9,2 |
| Actinium         | Thorium           | Protactinium     | Uranium          | Neptunium        | Plutonium        | Americium        | Curium           | Berkelium        | Californium      | Einsteinium      | Fermium          | Mendelevium      | Nobelium         | Lawrencium       |